

S&P 500 Volatility Forecasting

Executive Summary

Financial markets exhibit volatility clustering. Accurate volatility forecasts are essential for portfolio risk management, Value-at-Risk (VaR) calculations, derivative pricing, position sizing, and capital allocation.

This project develops a complete volatility forecasting pipeline using ten years of S&P 500 market data (2016–2026).

The study evaluates whether conditional volatility models can effectively capture market risk dynamics and compares the forecasting performance of GARCH(1,1), EWMA (RiskMetrics), and Rolling Volatility models.

Business Problem

Institutional investors are generally less concerned with predicting exact market direction and more concerned with forecasting future market risk.

Questions addressed:

- How volatile is the market expected to be tomorrow?
- Does volatility exhibit persistence over time?
- Can advanced volatility models outperform simple historical approaches?
- Which model provides the most accurate out-of-sample volatility forecasts?

These questions are directly relevant to risk management teams, quantitative research groups, hedge funds, and investment banks.

Dataset

- Source: Yahoo Finance
- Instrument: S&P 500 Index (^GSPC)
- Period: June 2016 – June 2026
- Frequency: Daily
- Observations: 2,513 trading days

Variables available:

- Open
- High
- Low
- Close
- Volume

For volatility modeling, only adjusted closing prices were required.

Methodology

Phase 1: Data Preparation

Daily closing prices were collected and transformed into logarithmic returns.

Logarithmic returns were selected because:

- Returns become additive across time.
- Statistical properties improve.
- Financial models typically assume returns rather than prices.

Phase 2: Exploratory Data Analysis

Three key visualizations were examined:

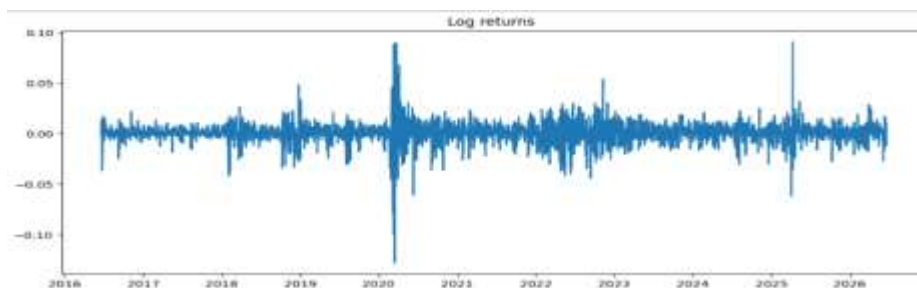
Price Series

The S&P 500 exhibited a strong upward trend and changing variance over time.



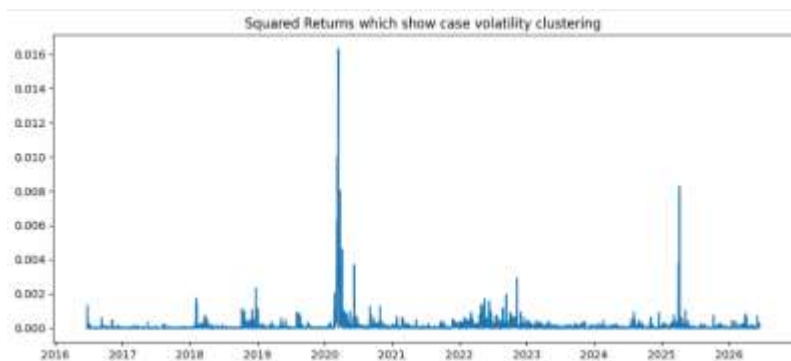
Return Series

Returns fluctuated around zero with no obvious trend.



Squared Returns

Periods of elevated volatility were observed to cluster together, indicating potential conditional heteroskedasticity.



This provided the first visual evidence of volatility clustering.

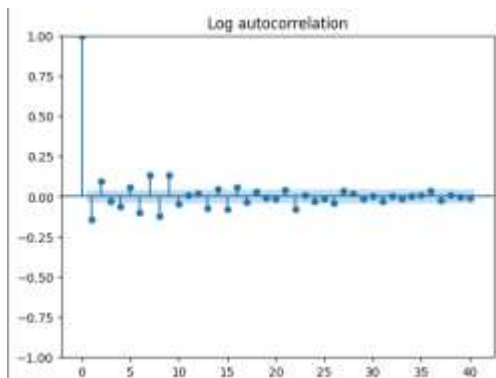
Phase 3: Stationarity Validation

An Augmented Dickey-Fuller (ADF) test was applied to the return series. The null hypothesis of a unit root was strongly rejected. S&P 500 returns are stationary and suitable for volatility modeling.

Phase 4: Volatility Clustering Analysis

Autocorrelation functions (ACF) were analyzed.

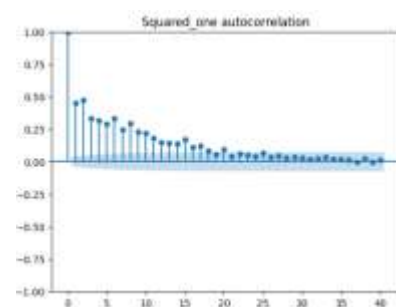
ACF of Returns :- Autocorrelations were close to zero.



Interpretation: Past returns contain little predictive information about future returns.

ACF of Squared Returns

Significant positive autocorrelation was observed.



Interpretation: Volatility exhibits persistence. This is a characteristic signature of financial time series and motivates the use of ARCH/GARCH model

Phase 5: ARCH Effect Testing

An ARCH-LM test was performed to formally test for conditional heteroskedasticity.

Strong ARCH effects were detected.

Volatility is time-dependent and suitable for GARCH modeling.

Phase 6: GARCH(1,1) Model Estimation

The following model was estimated:

$$\sigma^2_t = \omega + \alpha \varepsilon^2_{t-1} + \beta \sigma^2_{t-1}$$

Estimated Parameters:

$$\omega = 0.0371$$

$$\alpha = 0.1647$$

$$\beta = 0.8052$$

Persistence:

$$\alpha + \beta = 0.9699$$

Interpretation:

The market exhibits strong volatility persistence. Large shocks continue influencing volatility for extended periods before gradually decaying.

Phase 7: Residual Diagnostics

Model adequacy was assessed using standardized residuals.

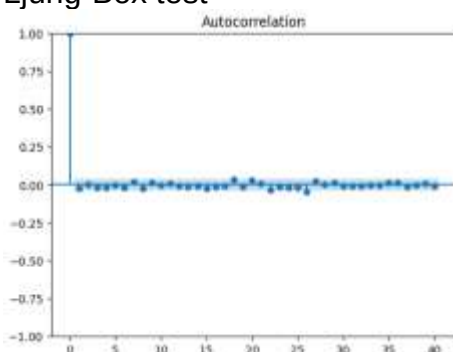
Tests Performed:

- ACF of standardized residuals
- ACF of squared standardized residuals
- Ljung-Box test

Conclusion:

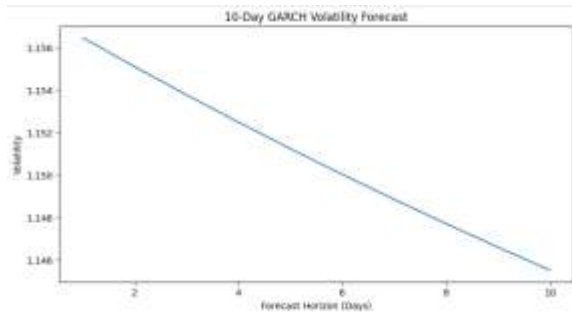
No significant autocorrelation remained after fitting the GARCH model.

The model successfully captured the volatility structure present in the data.



Phase 8: Volatility Forecasting

A 10-day volatility forecast was generated.



This behavior is consistent with the estimated persistence parameter, suggesting that volatility shocks decay over time and converge toward a long-run average.

Forecast results indicated a gradual decline in future volatility levels.

Phase 9: Benchmark Comparison

Three volatility estimation approaches were evaluated:

Rolling Volatility

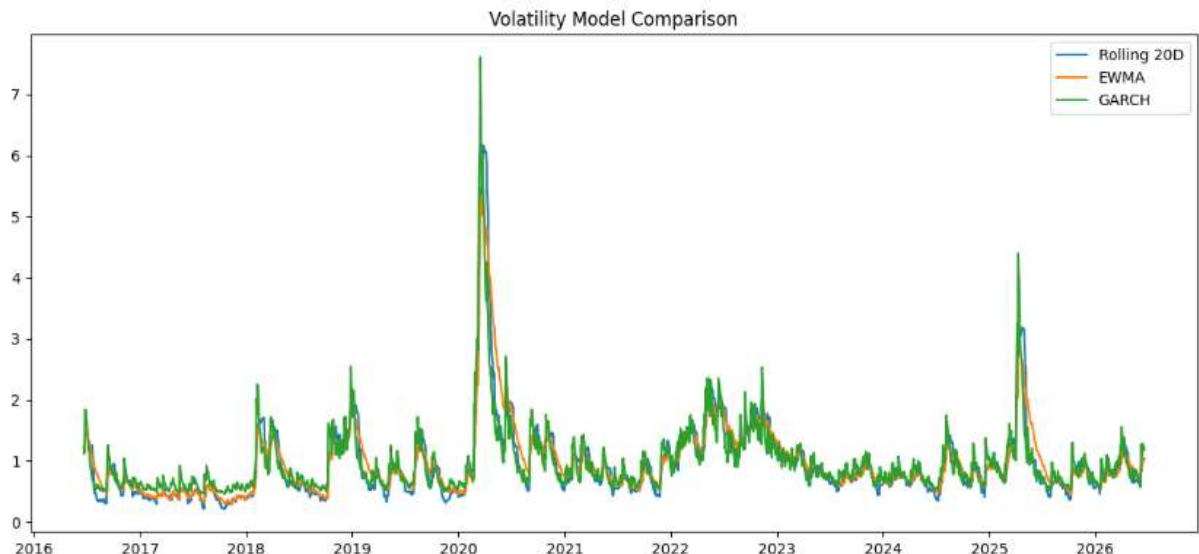
- Simple moving standard deviation
- Equal weight assigned to historical observations

EWMA (RiskMetrics)

- Exponentially weighted volatility estimate
- Greater emphasis on recent market movements

GARCH(1,1)

- Models both volatility persistence and new shocks
- Data-driven parameter estimation



Visual comparison showed **all models captured major market stress periods** including:

- COVID-19 crash
- 2022 market turbulence
- 2025 volatility spikes

GARCH produced the strongest response during high-volatility regimes.

Phase 10: Out-of-Sample Backtesting

Training Period:2016–2024

Interpretation:

Testing Period:2025–2026

Forecast accuracy was evaluated using RMSE.

EWMA provided the best forecasting performance during the evaluation period, while GARCH outperformed the traditional rolling volatility benchmark.

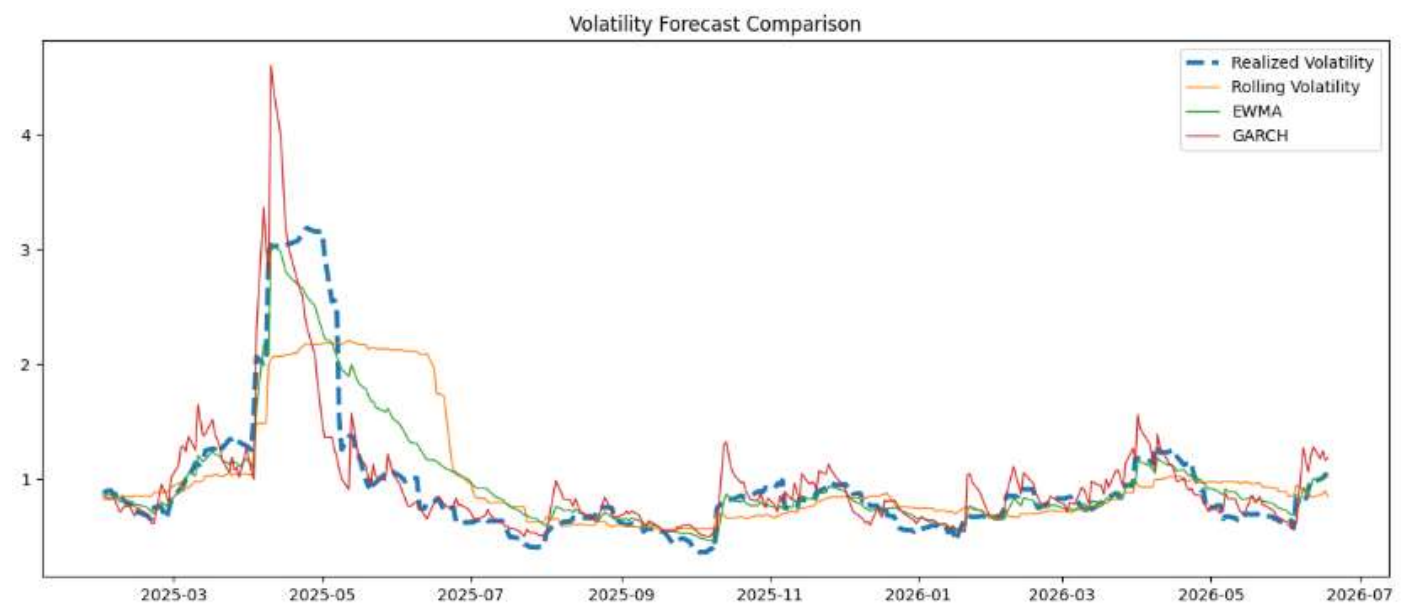
EWMA RMSE = 0.2347

GARCH RMSE = 0.3125

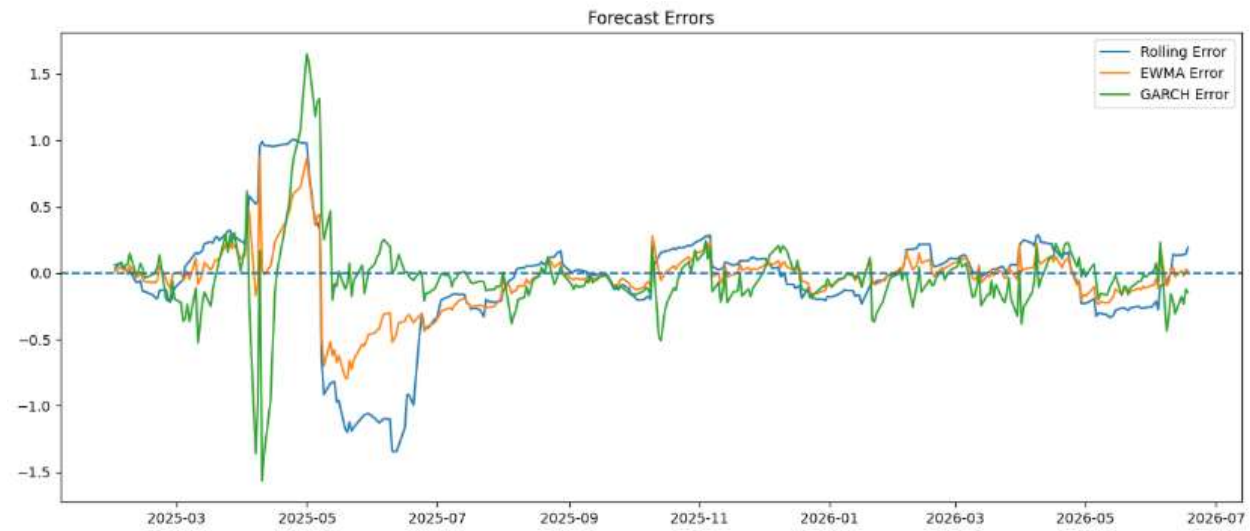
Rolling Volatility RMSE = 0.4201

This demonstrates that more complex models do not always guarantee superior predictive performance.

Here is the forecasting result of all models , tested on timeline of nearly one year 25-26.



Lastly the errors



Key Findings

1. S&P 500 returns are stationary.
 2. Significant volatility clustering exists.
 3. ARCH effects are statistically significant.
 4. GARCH(1,1) successfully models conditional volatility.
 5. Diagnostic tests indicate adequate model specification.
 6. EWMA achieved the best out-of-sample forecasting performance.
 7. Volatility persistence remains a defining characteristic of equity markets.
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Business Impact

Accurate volatility forecasts are critical for:

- Value-at-Risk (VaR) estimation
- Portfolio risk management
- Capital allocation
- Derivatives pricing
- Position sizing
- Stress testing

This project demonstrates how quantitative volatility models can transform raw market data into actionable risk estimates used by financial institutions.